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Definition 0.1. (Stochastic process) Let T be an ordered set, $(\Omega, \mathcal{F}, \mathbb{P})$ a probability space, and $(E, \mathcal{G})$ a measurable space. A stochastic process is a collection of random variables $X = \{X_t : t \in T\}$ such that for each fixed $t \in T$, X_t is a random variable from $(\Omega, \mathcal{F}, \mathbb{P})$ to $(E, \mathcal{G})$. The set Ω is known as the sample space, where E is the state space of the stochastic process X_t.

A stochastic process X may be viewed as a function of both $t \in T$ and $\omega \in \Omega$. We will sometimes write $X(t), X(t, \omega)$ or $X_t(\omega)$. For a fixed sample point $\omega \in \Omega$, the function $X_t(\omega) : T \to E$ is called a sample path(realisation, trajectory) of the process of X.

Definition 0.2. (finite-dimensional distributions) The finite-dimensional distributions(FDDs) of a stochastic process are the distributions of the E^k-valued random variables $(X(t_1), X(t_2), \ldots, X(t_k))$ for an arbitrary positive integer k and aribitrary times $t_i \in T$, $i \in \{1, \ldots, k\}$:

$$F(\boldsymbol{x}) = \mathbb{P}(X(t_i) \leq x_i,\ i = 1, \ldots, k),$$

with $\boldsymbol{x} = (x_1, \ldots x_k)$.

Definition 0.3. A one-dimensional continuous-time Gaussian process is a stochastic process for which $E = \mathbb{R}$ and all the FDDs are Gaussian, i.e., every finite-dimensional vector $(X_{t_1}, X_{t_2}, \ldots, X_{t_k})$ is an $\mathcal{N}(\mu_k, K_k)$ random variable for some vector μ_k and a symmetric non-negative definite matrix K_k for all $k = 1, 2, \ldots$ and for all $t_1, t_2, \ldots, t_k$.

Definition 0.4. A stochastic process is called (strictly) stationary if all FDDs are invariant under time translation: for every integer k and for all times $t_i \in T$, the distribution of $(X(t_1), X(t_2), \ldots, X(t_k))$ is equal to that of $(X(s+t_1), X(s+t_2), \ldots, X(s+t_k))$ for every s such that $s + t_i \in T$ for all $i \in \{1, \ldots, k\}$. In other words,

$$\mathbb{P}(X_{(t_1+s)} \in A_1, \ldots, X_{(t_k+s)} \in A_k) = \mathbb{P}(X_{t_1} \in A_1, \ldots, X_{t_k} \in A_k),\ s \in T.$$

Definition 0.5. A stochastic process $X_t \in L^2$ is called second-order stationary or weakly stationary if the first moment $\mathbb{E}X_t$ is a constant and the covariance function $\mathbb{E}(X_t - \mu)(X_s - \mu)$ depends only on the difference $t - s$:

$$\mathbb{E}X_t = \mu, \qquad \mathbb{E}((X_t - \mu)(X_s - \mu)) = C(t - s).$$

Definition 0.6. A one-dimensional standard *Brownian motion* $W(t) : \mathbb{R}^+ \to \mathbb{R}$ is a real-valued stochastic process with almost surely continuous paths such that $W(0) = 0$, it has independent increments, and for every $t > s \geq 0$, the increment $W(t) - W(s)$ has a Gaussian distribution with mean 0 and variance $t - s$. A standard d-dimensional standard Brownian motion $W(t) : \mathbb{R}^+ \to \mathbb{R}^d$ is a vector of d independent one-dimensional Brownian motions:

$$W(t) = (W_1(t), W_2(t), \dots, W_d(t)),$$

where $W_i(t), i = 1, \dots, d$, are independent one-dimensional Brownian motions.

The Ornstein-Uhlenbeck Process

A real-valued Gaussian stationary process defined on $\mathbb{R}$ with correlation function given by the following is called a stationary *Ornstein-Uhlenbeck process*

$$C(t) = C(0)e^{-\alpha|t|}, \qquad \alpha > 0$$

where $C(0) = \frac{D}{\alpha}$ with $D > 0$.

Diffusion process

Roughly speaking, a Markov process is a stochastic process that retains no memory of where it has been in the past: only the current state of a Markov process can influence where it will go next. More precisely, a Markov process is a stochastic process whose past and future are statistically independent, conditioned on its present state.
We define a continuous-time Markov process with state space $\mathbb{R}$ as a stochastic process whose future depends on its present state and not on how it got there:

$$\mathbb{P}(X_{t+h} \in \Gamma | \{X_s, s \leq t\}) = \mathbb{P}(X_{t+h} | X_t)$$

for all Borel sets Γ. We will only consider the continuous-time Markov process for which a *conditional probability* density exists:

$$\mathbb{P}(X_{t+h} \in \Gamma | X_t) = \int_\Gamma p(y, t+h | x, t) dy. \tag{1}$$

The Markov property enables us to obtain an evolution equation for the transition probability density defined in (1). This equation is the Chapman-Kolmogorov equation. Using this equation, we can study the evolution of a Markov process.

Example 0.7. Brownian motion is a Markov process with conditional probability density given by the following formula:

$$\mathbb{P}(W_{t+h} \in \Gamma | W_t = x) = \int_\Gamma \frac{1}{\sqrt{2\pi h}} \exp\left(-\frac{|x-y|^2}{2h}\right) \tag{2}$$

The Markov property of Brownian motion follows from the fact that it has independent increments.

Example 0.8. The stationary Ornstein-Uhlenbeck process $V_t = e^{-t}W(e^{2t})$ is a Markov process with conditional probability density

$$p(y,t|x,s) = \frac{1}{\sqrt{2\pi(1-e^{-2(t-s)})}} \exp\left(-\frac{|y-xe^{-(t-s)^2}|}{2(1-e^{-2(t-s)})}\right) \tag{3}$$

We will be concerned mostly with time-homogeneous Markov processes,i.e., processes for which the conditional probabilities are invariant under time shifts. Consider the case a continuous-time Markov process with continuous state space and with continuous paths. As we have seen in above two examples, Brownian motion is such a process. The conditional probability density of the Brownian motion (2) is the fundamental solution of the diffusion equation:

$$\partial_t p = \frac{1}{2}\partial_{yy} p, \quad \lim_{t\to s} p(y,t|x,s) = \delta(y-x). \tag{4}$$

Similarly, the conditional distribution of the Ornstein-Uhlenbeck process satisfies the initial value problem

$$\partial_t p = \partial_y(yp) + \frac{1}{2}\partial_{yy} p, \quad \lim_{t\to s} p(y,t|x,s) = \delta(y-x). \tag{5}$$

Brownian motion and the Ornstein-Uhlenbeck process are examples of a diffusion process: a continuous-time Markov process with continuous paths.

Markov process

The filtration generated by our stochastic process X_t, where X_t is :

$$\mathcal{F}_t^X := \sigma(X_s : s \le t).$$

Definition 0.9. Let X_t be a stochastic process defined on a probability space $(\Omega, \mathcal{F}, \mu)$ with values in $\mathbb{R}^d$, and let $\mathcal{F}_\sqcup{}^X$ be the filtration generated by $\{X_t : t \in \mathbb{R}_+\}$. Then $\{X_t : t \in \mathbb{R}_+\}$ is a Markov process if

$$\mathbb{P}(X_t \in \Gamma | \mathcal{F}_t^X) = \mathbb{P}(X_t \in \Gamma | X_s)$$

for all $t, s \in T$ with $t \ge s$, and $\Gamma \in \mathcal{B}(\mathbb{R}^d)$.

Chapman-Kolmogorov equation

With every continuous-time Markov process X_t defined in a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and state space $(\mathbb{R}^d, \mathcal{B})$, we can associate the *transition function*

$$P(\Gamma, t|X_s, s) := \mathbb{P}[X_t \in \Gamma|\mathcal{F}_s^X],$$

for all $t, s \in \mathbb{R}_+$ with $t \geq s$ and all $\Gamma \in \mathcal{B}(\mathbb{R}^d)$. It is a function of four arguments: the initial time s and position X_s, and the final time t and the set Γ. The transition function $P(\Gamma, t|x, s)$ is, for fixed t, x, s, a probability measure on $\mathbb{R}^d$ with $P(\mathbb{R}^d, t|x, s) = 1$; it is $\mathbb{R}^d$-measurable in x, for fixed t, s, Γ, and satisfies the Chapman-Kolmogorov equation

$$P(\Gamma, t|x, s) = \int_{\mathbb{R}^d} P(\Gamma, t|y, u)P(dy, u|x, s)$$

for all $x \in \mathbb{R}^d$, $\Gamma \in \mathcal{B}(\mathbb{R}^d)$, and $s, u, t \in \mathbb{R}_+$ with $s \leq u \leq t$. Assume that $X_S = x$. Since $\mathbb{P}(X_t \in \Gamma|\mathcal{F}_s^X) = \mathbb{P}(X_t \in \Gamma|X_S)$, we can write

$$P(\Gamma, t|x, s) = \mathbb{P}(X_t \in \Gamma|X_s = x).$$

We consider Markov processes whose transition function has a density with respect to the Lebesgue measure:

$$P(\Gamma, t|x, s) = \int_\Gamma p(y, t|x, s)dy$$

We will refer to $p(y, t|x, s)$ as the transition probability density. It is a function of four arguments: the initial position and time x, s, and the final position and time y, t. The Chapman-Kolmogorov equation becomes

$$\int_\Gamma p(y, t|x, s)dy = \int_{\mathbb{R}^d} \int_\Gamma p(y, t|z, u)p(z, u|x, s)dydz,$$

For time-homogeneous Markov processes we can fix the initial time, s=0. The Chapman-Kolmogorov equation for a time-homogeneous Markov process then becomes

$$P(t + s, x, \Gamma) := P(\Gamma, t|x, s) = P(\Gamma, t - s|x, 0) = \int_{\mathbb{R}^d} P(s, x, dz)P(t, z, \Gamma). \tag{6}$$

Given the initial distribtuion ν and the transition function $P(x, t, \Gamma)$ of a Markov process X_t , we can calculate the probability of finding X_t in a set Γ at time t:

$$\mathbb{P}(X_t \in \Gamma) = \int_{\mathbb{R}^d} P(x, t, \Gamma)\nu(dx)$$

The Generator of a Markov Process

Let X_t denote a time-homogeneous Markov process. The Chapman-Kolmogorov equation (6) suggests that a time-homogeneous Markov process can be described through a semigroup of operators,i.e., a one-parameter family of linear operators with the properties

$$P_0 = I \qquad P_{t+s} = P_t \circ P_s \qquad \forall t, s \geq 0$$

Indeed, let $P(t,\cdot,\cdot)$ be the transition function of a homogeneous Markov process and let $f \in C_b(\mathbb{R}^d)$, the space of continuous bounded functions on $\mathbb{R}^d$, and defined the operator:

$$(P_t f)(x) := \mathbb{E}(f(X_t)|X_0 = x) = \int_{\mathbb{R}^d} f(y)P(t,x,dy). \tag{7}$$

This is a linear operator with

$$P_0 f(x) = f(x),$$

which means that $P_0 = I$. Furthermore,

$$\begin{aligned} P_{t+s}(f)(x) &= \int_{\mathbb{R}^d} f(y)P(t+s,x,dy) = \int_{\mathbb{R}^d}\int_{\mathbb{R}^d} f(y)P(s,z,dy)P(t,x,dz) \\ &= \int_{\mathbb{R}^d} \Big(\int_{\mathbb{R}^d} f(y)P(s,z,dy)\Big)P(t,x,dz) = \int_{\mathbb{R}^d} (P_s f)(z)P(t,x,dz) \\ &= (P_t \circ P_s f)(x). \end{aligned}$$

Consequently,

$$P_{t+s} = P_t \circ P_s$$

Let now X_t be a time-homogeneous Markov process in $\mathbb{R}^d$, and let P_t denote the corresponding semigroup defined in (7). We consider this semigroup acting on continuous bounded functions and assume that $P_t f$ is also a $C_b(\mathbb{R}^d)$ function. We defined by $\mathcal{D}(\mathcal{L})$ the set of all $f \in C_b(E)$ such that the strong limit

$$\mathcal{L}f := \lim_{t\to 0} \frac{P_t f - f}{t} \tag{8}$$

exsits. The operator $\mathcal{L} : \mathcal{D}(\mathcal{L}) \to C_b(\mathbb{R}^d)$ is called the infinitesimal generator of the operator semigroup P_t. We will also refer to $\mathcal{L}$ as the generator of the Markov process X_t. The semigroup property and the definition of the generator of a Markov semigroup (8) imply that formally, we can write

$$P_t = e^{t\mathcal{L}}.$$

Consider the function $u(x,t) := (P_t f)(x) = \mathbb{E}(f(X_t)|X_0 = x)$. We calculate its time derivative:

$$\begin{aligned} \partial_t u &= \frac{d}{dt}(P_t f) = \frac{d}{dt}\big(e^{t\mathcal{L}} f\big) \\ &= \mathcal{L}\big(e^{t\mathcal{L}} f\big) = \mathcal{L}P_t f = \mathcal{L}u. \end{aligned}$$

Furthermore, $u(x,0) = P_0 f(x) = f(x)$. Consequently, $u(x,t)$ satisfies the initial value problem

$$\partial_t u = \mathcal{L}u, \tag{9}$$
$$u(x,0) = f(x). \tag{10}$$

Equation (10) is the *backward Kolmogorov equation.* It governs the evolution of the expectation of an observable $f \in C_b(\mathbb{R}^d)$. At this level, this is formal, since we do not have a formula for the generator $\mathcal{L}$ of the Markov semigroup. When the Markov process is the solution of a stochastic differential equation, then the generator is a second-order elliptic differential operator, and the backward Kolmogorov equation becomes an initial value problem for a parabolic partial differential equation.

The Adjoint Semigroup

The semigroup P_t acts on bounded continuous functions. We can also define the adjoint semigroup P_t^*, which acts on probability measures:

$$P_t^* \mu(\Gamma) = \int_{\mathbb{R}^d} \mathbb{P}(X_t \in \Gamma | X_0 = x) d\mu(x) = \int_{\mathbb{R}^d} P(t, x, \Gamma) d\mu(x). \tag{11}$$

The image of a probability measure μ under P_t^* is again a probability measure. The operators P_t and P_t^* are (formally) adjoint in the L^2-sense:

$$\int_{\mathbb{R}} P_t f(x) d\mu(x) = \int_{\mathbb{R}} f(x) d(P_t^* \mu)(x). \tag{12}$$

We can write

$$P_t^* = e^{t\mathcal{L}^*},$$

where $\mathcal{L}^*$is the L^2-adjoint of the generator of the process:

$$\int \mathcal{L} f h dx = \int f \mathcal{L}^* h dx$$

Let X_t be a Markov process with generator P_t with $X_0 \sim \mu$, and let P_t^* denote the adjoint semigroup defined in (11). We define

$$\mu_t := P_t^* \mu.$$

This is the *law* of the Markov process. An argument similar to that used in the derivation of the backward Kolmogorov equation (10) enables us to obtain an equation for the evolution of μ_t:

$$\partial_t \mu_t = \mathcal{L}^* \mu_t, \ \mu_0 = \mu \tag{13}$$

Assuming that the initial distribution μ and the law of the process μ_t each have a density with respect to Lebesgue measure, denoted by $\rho_0(\cdot)$ and $\rho(\cdot, t)$ respectively, this equation becomes

$$\partial_t \rho = \mathcal{L}^* \rho, \quad \rho(y, 0) = \rho_0(y). \tag{14}$$

This is the *forward Kolmogorov equation.* When the Markov process starts at $X_0 = x$, deterministic, the initial for the Fokker-Planck equation becomes $\rho_0 = \delta(x - y)$. As with the backward Kolmogorov equation (10), this equation is still formal, since we do not have a formula for the adjoint $\mathcal{L}^*$ of generator of the Markov process X_t.

Diffusion Process

Definition 0.10. A Markov process X_t in $\mathbb{R}$ with transition function $P(\Gamma, t|x, s)$ is called a diffusion process if the following conditions are satisfied:

1. **(Continuity)** For every x and every $\epsilon > 0$,

$$\int_{|x-y|>\epsilon} P(dy, t|x, s) = o(t - s), \tag{15}$$

uniformly over $s < t$.

2. **(Definition of drift coefficient)** There exists a function $b(x, s)$ such that for every x and every $\epsilon > 0$,

$$\int_{|x-y|\le\epsilon} (y - x)P(dy, t|x, s) = b(x, s)(t - s) + o(t - s), \tag{16}$$

uniformly over $s < t$.

3. **(Definition of diffusion coefficient)** There exists a function $\Sigma(x, s)$ such that for every x and every $\epsilon > 0$,

$$\int_{|x-y|\le\epsilon} (y - x)^2 P(dy, t|x, s) = \Sigma(x, s)(t - s) + o(t - s), \tag{17}$$

uniformly over $s < t$.

Theorem 0.11. (Kolmogorov) Assume that the stochastic diffusion process X_t is Markov and that $p(y, t|\cdot, \cdot), b(y, t), \Sigma(y, t)$ are smooth functions of y, t. Then the transition probability density is the solution to the initial value problem

$$\partial_t p = -\partial_y (b(y, t)p) + \frac{1}{2}\partial_{yy}(\Sigma(y, t)p), \quad p(y, 0|x, 0) = \delta(y - x). \tag{18}$$

Assume now that the initial distribution of X_t is $\rho_0(x)$, and set $s = 0$ in (18). Define

$$p(y, t) = \int p(y, t|x, 0)\rho_0(x)dx. \tag{19}$$

We multiply the forward Kolmogorov equation (18) by $\rho_0(x)$ and integrate with respect to x to obtain the equation

$$\partial_t p(y, t) = -\partial_t (b(y, t)p(y, t)) + \frac{1}{2}\partial_{yy}(\Sigma(y, t)p(y, t)), \tag{20}$$

together with the initial condition

$$p(y, 0) = \rho_0(y) \tag{21}$$

1 SDE

We consider SDEs of the form

$$\frac{dX(t)}{dt} = b(X(t)) + \sigma(X(t))\xi(t), \qquad X(0) = x, \tag{22}$$

where $X(t) \in \mathbb{R}^d$, $b : \mathbb{R}^d \to \mathbb{R}^d$, and $\sigma : \mathbb{R}^d \to \mathbb{R}^{d\times m}$. we use the notation $\xi(t) = \frac{dW}{dt}$ to denote (formally) the derivative of Brownian motion in $\mathbb{R}^m$, i.e., the white noise process, which is a (generalised) mean-zero Gaussian vector-value stochastic process with covariance function

$$\mathbb{E}(\xi_i(t)\xi_j(s) = \delta_{ij}\delta(t-s), \quad i, j = 1, 2, \dots, m.$$

The initial condition x can be either deterministic or a random variable that is independent of the Brownian motion $W(t)$.

The amplitude of the noise in (22) may be independent of the state of the system, $\sigma(x) \equiv \sigma$, a constant, in which case we will say that the noise in (22) is *additive*. When the amplitude of the noise depends on the sate of the system, we will say that the noise in (22) is *multiplicative*.

Since the white noise process $\xi(t)$ is defined only in a generalised sense, Eq.(22) is only formal. We will usually write it in the form

$$dX(t) = b(X(t))dt + \sigma(X(t))dW(t), \tag{23}$$

together with the initial condition $X(0) = x$. In fact the correct interpretation of the SDE (23) is as a stochastic integral equation

$$X(t) = x + \int_0^t b(X(t))dt + \int_0^t \sigma(X(t))dW(t) \tag{24}$$

The Ito and Stratonovich Stochastic Integrals

We will define stochastic integrals of the form

$$I(t) = \int_0^t f(s)dW(s), \tag{25}$$

where $W(t)$ is a standard one-dimensional Brownian motion and $t \in [0, T]$. Our goal is to define the stochastic integral $I(t)$ as the L^2-limit of a Riemann sum approximation of (25). To this end, we introduce a partition of the interval $[0, T]$ by setting $t_k = k\Delta t$, $k = 0, \dots, K-1$, and $K\Delta t = t$; we also define a parameter $\lambda \in [0, 1]$ and set

$$\tau_k = (1-\lambda)t_k + \lambda t_{k+1}, \qquad k = 0, 1, \dots, K-1. \tag{26}$$

We define now the stochastic integral as the $L^2(\Omega)$ limit (Ω denoting the underlying probability space) of the Riemann sum approximation

$$I(t) := \lim_{K\to\infty} \sum_{k=0}^{K-1} f(\tau_k)(W(t_{k+1}) - W(t_k)). \tag{27}$$

In contrast to the case of the Riemann-Stieltjes integral, when we integrate against a smooth deterministic function, the result in (27) depends on the choice of $\lambda \in [0,1]$ in (26). The two most common choices are $\lambda = 0$, in which case we obtain the *Ito stochastic integral*

$$I_I(t) := \lim_{K \to \infty} \sum_{k=0}^{K-1} f(\tau_k)(W(t_{k+1}) - W(t_k)), \tag{28}$$

and $\lambda = \frac{1}{2}$, which leads to the *Stratonovich stochastic integral*

$$I_S(t) := \lim_{K \to \infty} \sum_{k=0}^{K-1} f\big(\frac{1}{2}(t_k + t_{k+1})\big)(W(t_{k+1}) - W(t_k)). \tag{29}$$

We will use the notation

$$I_S(t) = \int_0^t f(s) \circ dW(s)$$

to denote the stratonovich stochastic integral. In general, the Ito and Stratonovich stochastic integrals are different. When the integrand $f(t)$ depends on the Brownian motion $W(t)$ through $X(t)$, the solution of the SDE in (22), a formula exists for converting one stochastic integral into another. When the integrand in (28) is a sufficiently smooth function, then the stochastic integral is independent of the parameter λ, and in particular, the Ito and Stratonovich stochastic integrals coincide.

The Ito stochastic integral $I(t)$ is continuous almost surely in t. As expected, it satisfies the linearity property

$$\int_0^T \big(\alpha f(t) + \beta g(t)\big) dW(t) = \alpha \int_0^T f(t) dW(t) + \beta \int_0^T g(t) dW(t), \qquad \alpha, \beta \in \mathbb{R},$$

for all square-integrable functions $f(t), g(t)$. Furthermore, the Ito stochastic integral satisfies the Ito *isometry*

$$\mathbb{E}\Big(\int_0^T f(t) dW(t)\Big)^2 = \int_0^T \mathbb{E}|f(t)|^2 dt,$$

from which it follows that for all square-integrable functions f, g

$$\mathbb{E}\Big(\int_0^T h(t) dW(t) \int_0^T g(s) dW(s)\Big) = \mathbb{E} \int_0^T h(t) g(t) dt.$$

The Ito stochastic integral is a *martingale*:

Definition 1.1. Let $\{\mathcal{F}_\sqcup\}_{t \in [0,T]}$ be a filtration defined on the probability space $(\Omega, \mathcal{F}, \mu)$, and let $\{\mathcal{M}_t\}_{t \in [0,T]}$ be adapted to $\mathcal{F}_t$ with $\mathcal{M}_t \in L^1(0,T)$. We say that $\mathcal{M}_t$ is an $\mathcal{F}_t$ martingale if

$$\mathbb{E}[\mathcal{M}_t | \mathcal{F}_s] = \mathcal{M}_s, \quad \forall t \geq s.$$

For the Ito stochastic integral, we have

$$\mathbb{E} \int_0^T f(s) dW(s) = 0 \tag{30}$$

and

$$\mathbb{E}\Big[\int_0^T f(l)dW(l)|\mathcal{F}_s\Big] = \int_0^s f(l)dW(l) \quad \forall t \geq s, \tag{31}$$

where $\mathcal{F}_s$ denotes the filtration generated by $W(s)$.

Ito's formula

Consider the Ito SDE

$$dX_t = b(X_t)dt + \sigma(X_t)dW_t \tag{32}$$

where for simplicity, we have assumed that the coefficients are independent of time. Here X_t is a diffusion process with drift $b(x)$ and diffusion matrix

$$\Sigma(x) = \sigma(x)\sigma(x)^T.$$

The generator $\mathcal{L}$ is then defined as

$$\mathcal{L} = b(x) \cdot \nabla + \frac{1}{2}\Sigma(x) : D^2, \tag{33}$$

where D^2 denotes the Hessian matrix. The generator canalso be written as

$$\mathcal{L} = \sum_{j=1}^{d} b_j(x)\frac{\partial}{\partial x_j} + \frac{1}{2}\sum_{i,j=1}^{d} \frac{\partial^2}{\partial x_i \partial x_j}. \tag{34}$$

We can now write Ito formula using the generator. This formula enables us to calculate the rate of change in time of functions $V : [0,T] \times \mathbb{R}^d \to \mathbb{R}$ evaluated at the solution of an $\mathbb{R}^d$-valued SDE. First, recall that in the absence of noise, the rate of change of V can be written as

$$\frac{d}{dt}V(t, x(t)) = \frac{\partial V}{\partial t}(t, x(t)) + \phi V(t, x(t), \tag{35}$$

where $x(t)$ is the solution of the ODE $\dot{x} = b(x)$, and ϕ denotes the (backward) *Liouville operator*

$$\phi = b(x) \cdot \nabla. \tag{36}$$

Let now X_t be the solution of (32) and assume that the white noise W is replaced by a smooth function $\xi(t)$. Then the rate of change of $V(t, X_t)$ is given by the formula

$$\frac{d}{dt}V(t, X_t) = \frac{\partial V}{\partial t}(t, X_t) + \phi V(t, X_t) + \langle \nabla V(t, X_t), \sigma(X_t)\xi(t)\rangle \tag{37}$$

This formula is no longer valid when (32) is driven by white noise and not a smooth function. In particular, the Leibniz formula (37) has to be modified by the addition of a drift term that accounts for the lack of smoothness of the noise: the Liouville operator ϕ given by (36) in (37) has to be replaced by the generator $\mathcal{L}$ given by (33). Formally we can write

$$\frac{d}{dt}V(t, X_t) = \frac{\partial V}{\partial t}(t, X_t) + \mathcal{L}V(t, X_t) + \left\langle \nabla V(t, X_t), \sigma(X_t)\frac{dW}{dt} \right\rangle \tag{38}$$

The precise interpretation of the expression for the rate of change of V is in integrated form.

Theorem 1.2. (Ito's Formula) Assume that . Let X_t be the solution of (32), and let $V \in C^{1,2}([0,T] \times \mathbb{R}^d)$. Then the process $V(X_t)$ satisfies

$$V(t, X_t) = V(X_0) + \int_0^T \frac{\partial V}{\partial s}(s, X_s)ds + \int_0^T \mathcal{L}V(s, X_s)ds$$
$$+ \int_0^T \left\langle \nabla V(t, X_t), \sigma(X_t)\frac{dW}{dt} \right\rangle$$

derivation of the Fokker-Planck Equation

The Fokker-Planck operator corresponding to the SDE (32) reads

$$\mathcal{L}^* \cdot := \nabla \cdot \big(-b(x) \cdot + \frac{1}{2} \nabla \cdot (\Sigma \cdot) \big). \tag{39}$$

We can derive this formula from the formula for the generator $\mathcal{L}$ of X_t and two integration by parts:

$$\int_{\mathbb{R}^d} \mathcal{L}fh dx = \int_{\mathbb{R}^d} f \mathcal{L}^* h dx,$$

for all $f, h \in C_0^2(\mathbb{R}^d)$.

Proposition 1.3. Let X_t denote the solution of the Ito SDE (32) and assume that the initial condition X_0 is a random variable, independent of the Brownian motion driving the SDE, with density $\rho_0(x)$. Assume that the law of the Markov process X_t has a density $\rho(x,t) \in C^{2,1}(\mathbb{R}^d \times (0, +\infty))$. Then ρ is the solution of the initial value problem for the Fokker-Planck equation

$$\partial_t \rho = \mathcal{L}^* \rho \quad \forall (x,t) \in \mathbb{R}^d \times (0, \infty), \tag{40}$$

$$\rho = \rho_0 \qquad \forall x \in \mathbb{R}^d \times \{0\} \tag{41}$$

2 Examples of SDEs

Example 2.1. (Brownian Motion) We consider an SDE with no drift and constant diffusion coefficient:

$$dX_t = \sqrt{2\sigma} dW_t, \quad X_0 = x, \tag{42}$$

where the initial condition can be either deterministic or random, independent of the Brownian motion W_t. The solution is

$$X_t = x + \sqrt{2\sigma} W_t.$$

This is just Brownian motion starting at x with diffusion coefficient σ

Example 2.2. (Ornstein-Uhlenbeck Process) Adding a restoring force to (42), we obtain the SDE for the Ornstein-Uhlenbeck process that we have already encountered:

$$dX_t = \alpha X_t dt + \sqrt{2\sigma} dW_t, \quad X(0) = x, \tag{43}$$

with $\alpha > 0$ and where, as in the previous example, the initial condition can be either deterministic or random, independent of the Brownian motion W_t. We solve this equation using the variation of constant formula:

$$X_t = e^{-\alpha t} x + \sqrt{2\sigma} \int_0^t e^{-\alpha(t-s)} dW_s. \tag{44}$$

The generator is

$$\mathcal{L} = -\alpha x \frac{d}{dx} + \sigma \frac{d^2}{dx^2}.$$

2.1 Pictures

Figure 1: Sydney, NSW

2.2 Citation

This is a citation[1].

References

[1] H. Ren, "Template for math notes," 2021.